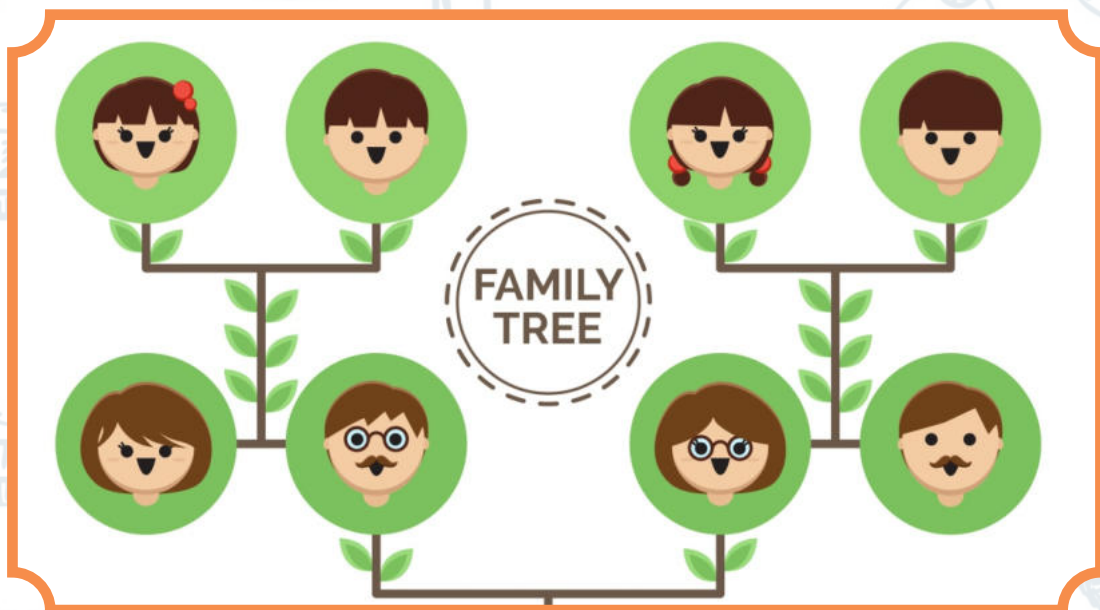


UNIT

23

INHERITANCE





Laws of Mendel

Introduction

- **Gregor John Mendel**, an Austrian monk, is known as the Father of Genetics.
- His success in genetic studies was due to his:
 - Careful choice of experimental material (*Pisum sativum* – garden pea)
 - Use of scientific methods and statistical analysis
 - Focus on clear, contrasting traits
 - Use of controlled cross-pollination and self-pollination

	Height	Seed Shape	Seed Color	Seed Coat Color	Pod Shape	Pod Color	Flower Position
Dominant	 Tall	 Round	 Yellow	 Green	 Inflated (full)	 Green	 Axial
Recessive Trait	 Short	 Wrinkled	 Green	 White	 Constricted (flat)	 Yellow	 Terminal

Reasons for Mendel's Success

- Chose ***Pisum sativum* (garden pea)** which was:
 - Easy to cultivate
 - Self-pollinating (but could be cross-pollinated manually)
 - Had clearly defined contrasting traits (e.g., tall vs. dwarf, round vs. wrinkled seeds)
 - Produced fertile offspring through cross-pollination
 - Avoided the biological complexities that confused earlier researchers by focusing on observable traits.

Significance

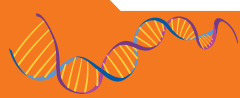
- Mendel's experiments laid the foundation of **modern genetics**.
- His methodical approach and choice of traits allowed him to deduce the fundamental **laws of inheritance**:

- **Law of Segregation**
- **Law of Independent Assortment**
- **Law of Dominance**

Mendel's Experimental Method

Mendel's Approach to Genetics

- **Started with pure-breeding plants:**
 - Mendel selected plants that consistently produced offspring with the same traits (pure lines).
- **Choose plants with contrasting characters:**
 - For example: smooth seeds vs. wrinkled seeds





Experimental Procedure

1. Selection of Traits:

- Chose **seven pairs of contrasting traits** such as:
- Seed shape: smooth vs. wrinkled
- Seed color: yellow vs. green
- Flower color: purple vs. white
- Pod shape: full vs. constricted
- Pod color: green vs. yellow
- Flower position: axial vs. terminal
- Plant height: tall vs. dwarf

2. Cross-Pollination:

- Manually transferred pollen from one plant to another with contrasting traits (e.g., from a smooth-seeded plant to a wrinkled-seeded plant).
- Ensured **controlled breeding** to observe inheritance patterns.

3. Observation of Offspring:

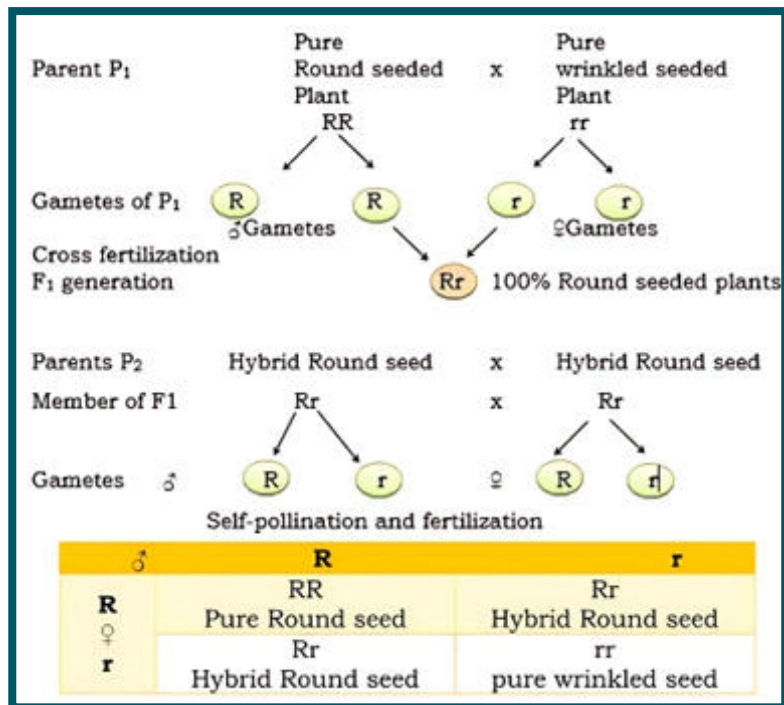
- Recorded the traits in the **first generation (F_1)** and **second generation (F_2)**.
- Noted consistent ratios in inheritance, forming the basis of genetic laws.

Purpose

- To **understand how traits are inherited** across generations.
- To develop a **scientific explanation** for inheritance patterns using controlled experiments.

S. No	Cross between contrasting characters	Results of 1 st Generation (F_1)	Results of 2 nd Generation (F_2)	Ratio in F_2
1	Tall x dwarf	All Tall	787 Tall 277 dwarf	2.84:1
2	Round seed x wrinkled seed	All Round seed	5474 Round 1850 wrinkled	2.96:1
3	Yellow cotyledon x green cotyledon	All Yellow cotyledon	6022 Yellow 2001 green	3.01:1
4	Purple flowers x white flowers	All Purple flower	705 Purple 224 white	3.15:1
5	Smooth pods x constricted pods	All Smooth pods	882 Smooth 299 constricted	2.95:1
6	Green pods x yellow pods	All Green pods	428 Green 152 yellow	2.82:1
7	Axial flower x terminal flower	All Axial flowers	651 Axial 207 terminal	3.14:1





Mendel's Results and Laws of Inheritance

Monohybrid Cross (Single Trait)

- **F₁ Generation** (First Filial):
- Cross: Pure tall (TT) × Pure dwarf (tt)
- All offspring were **tall**.
- Concluded: Trait for **tallness is dominant**, dwarfness is recessive.
- **F₂ Generation** (Second Filial):
- Self-crossing F₁ (Tt × Tt)
- Resulting ratio:
- **3 tall : 1 dwarf**
- **Genotypic ratio:** 1 TT : 2 Tt : 1 tt

Dihybrid Cross (Two Traits)

- **Cross example:** Yellow round (YYRR) × Green wrinkled (yyrr)
- **F₁ Generation:** All offspring were **yellow round** (dominant traits).
- **F₂ Generation:** Traits assorted independently.
- **Phenotypic ratio: 9:3:3:1**
- 9 yellow round

Law of Segregation (Mendel's First Law)

Definition:

- This law explains the inheritance of a **single trait** (monohybrid cross).
- It states that **each trait is controlled by a pair of factors** (genes) which **segregate** (separate) during gamete formation.
- As a result, **each gamete receives only one factor** (gene) from the pair.

Also Known As:

- **Law of Purity of Gametes** because gametes carry only one gene from the gene pair of a trait.





Important Terms (Modern Terminology vs Mendelian Terms)

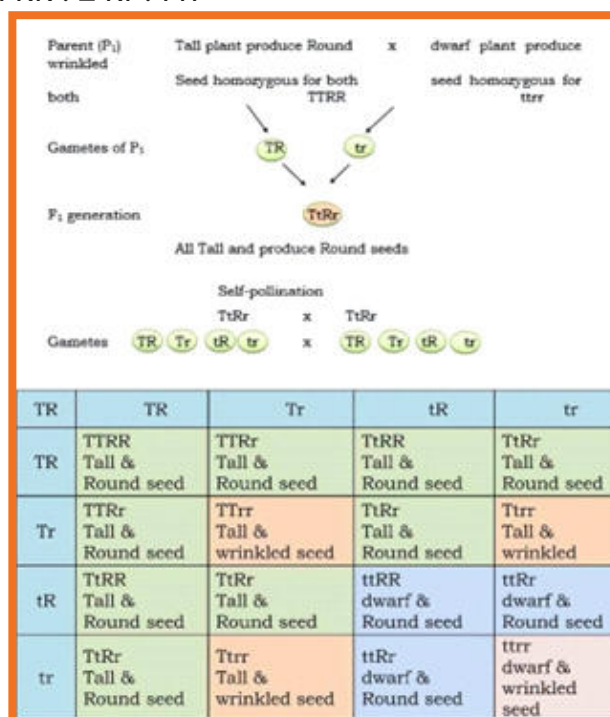
Mendelian Term	Modern Equivalent
Factor	Gene
Character	Trait
Pure breed	Homozygous (e.g., RR or rr)
Hybrid	Heterozygous (e.g., Rr)
Physical appearance	Phenotype
Gene pair (from parents)	Genotype

Monohybrid Cross Example (Round vs Wrinkled Seeds)

- **Parent Genotypes:**
- Round (dominant) = R
- Wrinkled (recessive) = r
- **Cross: Rr × Rr (Self-pollination of heterozygous plant)**

Genotype RR Rr rr
Phenotype Round Round Wrinkled

- **Phenotypic Ratio:** 3 Round : 1 Wrinkled
- **Genotypic Ratio:** 1 RR : 2 Rr : 1 rr



Test Cross

Purpose:

- To determine whether a phenotypically dominant individual is **homozygous (RR)** or **heterozygous (Rr)**.

Method:

- Cross the dominant individual with a **homozygous recessive (rr)** individual.

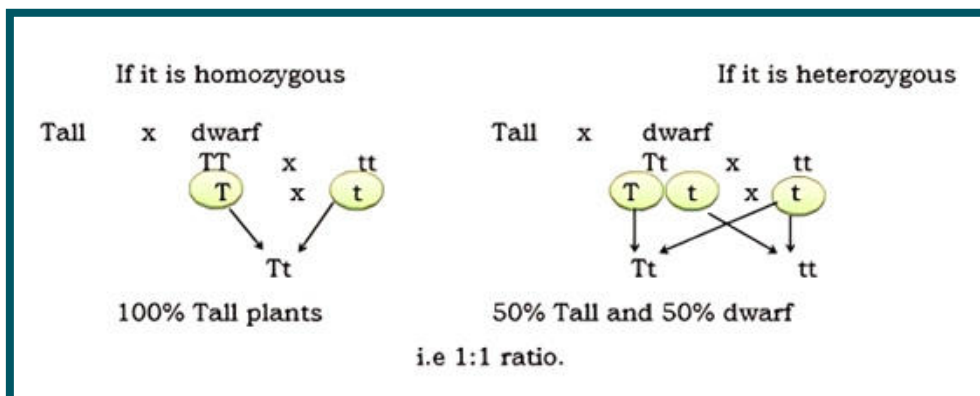




Outcomes:

Dominant Genotype	Cross with rr	Offspring Outcome
RR (Homozygous)	RR × rr	All Round (phenotypically dominant)
Rr (Heterozygous)	Rr × rr	1 Round : 1 Wrinkled

- 3 yellow wrinkled
- 3 green round
- 1 green wrinkled



Inheritance of Two Traits (Dihybrid Cross)

Introduction:

- Mendel studied not just **one trait** at a time but also how two **traits** are inherited together.
- This type of experiment is called a **Dihybrid Cross**.
- The result of this experiment is expressed as a **Dihybrid Ratio**.

Example of Mendel's Dihybrid Cross:

- **Parent 1:** Pure **Tall** plant with **Round** seeds (TTRR)
- **Parent 2:** Pure **Dwarf** plant with **Wrinkled** seeds (ttrr)
- **F₁ Generation:** All plants were **Tall** and had **Round** seeds (TtRr)

Self-Pollination of F₁ (TtRr × TtRr):

Gametes Produced:

- From each parent: TR, Tr, tR, tr

F₂ Generation (Punnett Square Result):

Phenotype	Combination Type
Tall, Round	Parental
Tall, Wrinkled	Non-Parental
Dwarf, Round	Non-Parental
Dwarf, Wrinkled	Parental





- **Phenotypic Ratio (Dihybrid Ratio):**

9:3:3:1

- 9 Tall & Round
- 3 Tall & Wrinkled
- 3 Dwarf & Round
- 1 Dwarf & Wrinkled

Law of Independent Assortment:

Definition:

- "The factors (genes) of different traits assort **independently** of each other during gamete formation."

Implication:

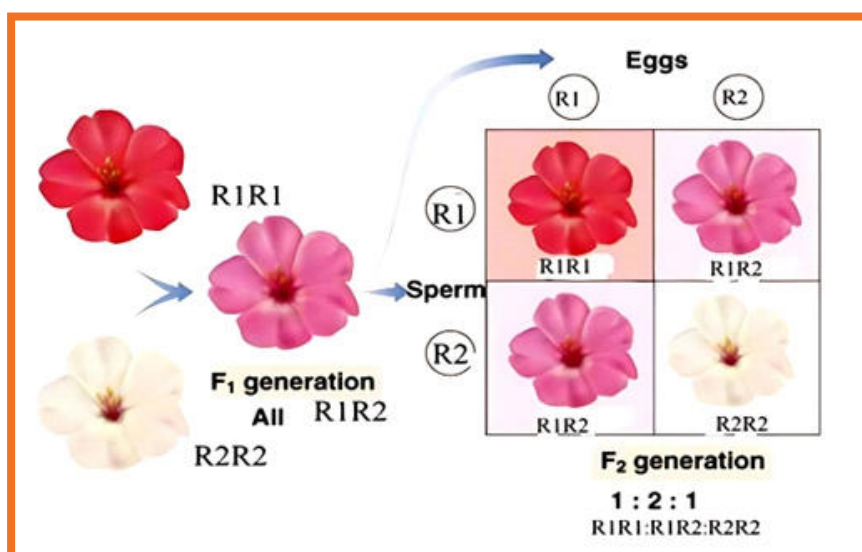
- Inheritance of one trait **does not influence** the inheritance of another trait.
- This explains how **non-parental combinations** (new combinations of traits) appear in F_2 generation.

Practice Exercise (Mendel's Seed Color Study)

Green (G) = dominant

Yellow (g) = recessive

Cross Type	Green Offspring	Yellow Offspring	Interpretation
i) Green $\times$ Yellow	82	78	Both likely heterozygous (Gg $\times$ gg)
ii) Green $\times$ Green	118	39	Likely Gg $\times$ Gg (F_2 ratio $\approx$ 3:1)
iii) White $\times$ Yellow	0	50	White likely recessive or unrelated
iv) Green $\times$ Yellow	74	0	Green likely homozygous dominant (GG)
v) Green $\times$ Green	90	0	Both likely homozygous dominant (GG)





Exceptions to Mendel's Laws of Inheritance

Background:

- **Gregor Mendel**, a monk and pioneer of genetics, formed basic principles of inheritance based on **pea plant experiments**.
- He proposed:
- Traits are controlled by **pairs of alleles**.
- One allele is dominant, and the other is **recessive**.
- When both alleles are present, **only the dominant allele is expressed** – known as **complete dominance**.

Limitations of Mendel's Laws:

- Mendel's work assumed:
- Only **two alleles per trait**.
- Clear **dominant–recessive relationships**.
- **Modern genetics** has found many **exceptions** to these rules.

Non-Mendelian Inheritance (Exceptions)

1. Incomplete Dominance

- Neither allele is completely dominant over the other.
- **Heterozygous phenotype** is an intermediate blend of both traits.
- **Example:** Red (RR) × White (WW) flowers → Pink (RW) flowers.

2. Co-Dominance

- Both alleles in a heterozygote are **fully expressed**.
- No blending; both traits appear simultaneously.
- **Example:** Blood group AB (both A and B alleles are expressed).

3. Multiple Alleles

- More than two alleles exist for a single gene in the population.
- **Example:** Human ABO blood group has three alleles – **I^A, I^B, i**.

4. Polygenic Inheritance (not in your text but commonly grouped with exceptions)

- A trait is controlled by multiple genes (not just one pair).
- Results in continuous variation.
- **Example:** Skin color, height in humans.

Incomplete Dominance

Definition:

- In **incomplete dominance**, **neither allele** is completely dominant over the other.
- In heterozygous individuals, **both alleles partially express**, resulting in a **blended phenotype**.

Discovery:

- **Discovered by:** Carl Correns in **1889**
- **Plant Studied:** 4 o'clock plant (*Mirabilis jalapa*)

Experiment Summary:

1. P (Parent) Generation:

- **Red-flowered plant** (homozygous) × **White-flowered plant** (homozygous)

2. F₁ Generation:

- All **Pink-flowered plants** (heterozygous)
- **No parental phenotype** (Red or White) observed





3. F₂ Generation (F₁ self-fertilized):

- Phenotypic ratio: 1 **Red** : 2 **Pink** : 1 **White**
- Genotypes:
- **Red** = R₁R₁ (homozygous)
- **Pink** = R₁R₂ (heterozygous)
- **White** = R₂R₂ (homozygous)

Key Features of Incomplete Dominance:

- **Blending of traits** in heterozygous condition
- **No masking** of one allele by the other
- Produces a **third, intermediate phenotype**
- **Genotypic ratio = Phenotypic ratio** in F₂ generation: 1:2:1

Allele Representation:

- Since no allele is completely dominant, alleles are denoted with **same letter and subscripts**:
- R₁ = Red flower allele
- R₂ = White flower allele
- R₁R₂ = Pink flower

Conclusion:

- Incomplete dominance is an **exception to Mendel's Law of Dominance**.
- It highlights a more **complex pattern of inheritance** where both alleles contribute to the phenotype.

Co-Dominance

Definition:

- **Co-dominance** is a form of **non-Mendelian inheritance** where **both alleles** in a **heterozygous** individual are **fully expressed**.
- The expression of alleles is **independent and distinct**, with **no blending or masking** of traits.

Key Features:

- Both **contrasting alleles** at the same locus express themselves.
- **No allele is dominant or recessive** over the other.
- Results in offspring that **simultaneously display traits** from both parents.

Example 1: Roan Coat in Cattle

- **Parent 1:** Red-coated cattle (homozygous)
- **Parent 2:** White-coated cattle (homozygous)
- **Offspring (F₁):** Roan-coated cattle (heterozygous)

The roan coat is not a blend of red and white, but a **mixture of red and white hairs** on the body—**both alleles are expressed independently**.

Example 2: MN Blood Group in Humans

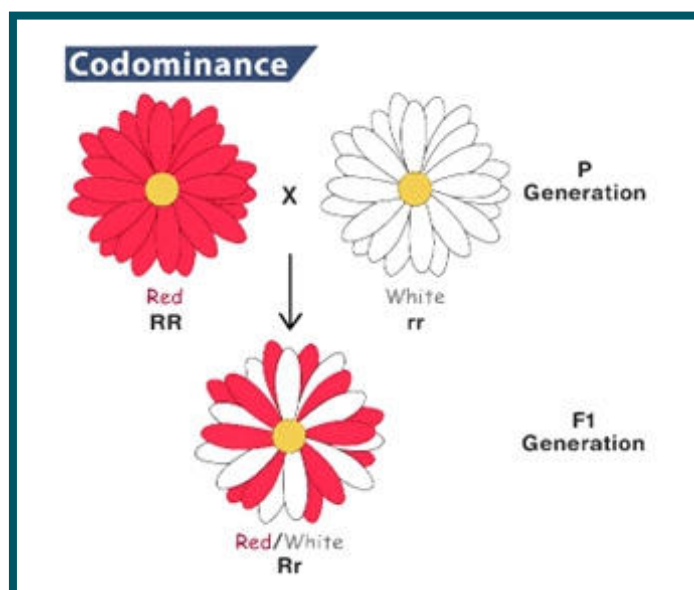
- Discovered by Landsteiner and Levine
- Based on presence of **M** and **N** antigens on red blood cells:
- **MM** genotype = M blood type
- **NN** genotype = N blood type
- **MN** genotype = **Both M and N antigens** expressed → **Co-dominant phenotype**





Conclusion:

- **Co-dominance differs from incomplete dominance:**
- In **incomplete dominance**, traits **blend**.
- In **co-dominance**, traits appear **side-by-side**.
- Co-dominance shows that **inheritance is more complex** than Mendel's original laws.



Comparison: Incomplete Dominance vs. Co-Dominance

Feature	Incomplete Dominance	Co-Dominance
Definition	Neither allele is completely dominant; results in blending	Both alleles are equally dominant and expressed together
Phenotype in Heterozygote	Intermediate/blended trait (mix of both alleles)	Both traits appear side by side, distinctly
Allele Interaction	Partial expression of both alleles	Full, independent expression of both alleles
Example (Plant)	Red × White → Pink (Mirabilis jalapa)	—
Example (Animal)	—	Red cow × White cow → Roan cow (red & white hairs visible)
Example (Human)	—	MN blood group: Both M and N antigens on red blood cells
F₂ Phenotypic Ratio	1 (Red) : 2 (Pink) : 1 (White)	1 (Red) : 2 (Roan) : 1 (White)
Blending Occurs?	Yes	No
Mendelian Exception?	Yes	Yes





Multiple Alleles

Definition:

- **Multiple alleles** refer to **more than two alternative forms** of a gene that can **occupy the same gene locus** on a chromosome.
- Although an organism can **only inherit two alleles** (one from each parent), **more than two alleles** may exist in the population due to **mutations**.

Key Characteristics:

- Multiple alleles are **different forms of the same gene**.
- Found at the **same locus** on homologous chromosomes.
- Each **diploid individual** carries **only two alleles**.
- Each **haploid gamete** contains **only one allele**.
- Some traits may involve **hundreds of possible alleles**.

Example: ABO Blood Group in Humans

Discovery:

- **Discovered by:** Karl Landsteiner (1901)
- **Genetic explanation by:** Bernstein (1924)

Genes Involved:

- Three alleles:
- I^A (codes for A antigen)
- I^B (codes for B antigen)
- i (no antigen; recessive)

Phenotypes and Genotypes:

Blood Group	Genotype(s)	Antigen(s) on RBC	Phenotype
A	$I^A I^A$ or $I^A i$	A	A (homozygous/heterozygous)
B	$I^B I^B$ or $I^B i$	B	B (homozygous/heterozygous)
AB	$i I^A I^B$	A and B	AB (co-dominant)
O	ii	None	O (recessive)

Summary:

- ABO blood group system demonstrates multiple alleles and co-dominance (I^A and I^B).
 - Blood group O is recessive, while A and B are dominant over i .
- I^A and I^B are co-dominant with respect to each other.

	GROUP A	GROUP B	GROUP AB	GROUP O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A & Anti-B
Antigens in red blood cell	A antigen 	B antigen 	A and B antigen 	None





Co-Dominance in Human Blood Group AB

Definition:

● **Co-dominance** is a type of **non-Mendelian inheritance** where **both contrasting alleles** at the same gene locus are **fully and independently expressed** in a heterozygous individual.

Example: Blood Group AB (ABO System)

- The **AB blood group** in humans demonstrates **co-dominance**.
- It results from the **IAIB** genotype.
- Both **IA (A antigen)** and **IB (B antigen)** are **equally expressed** on the **same red blood cell (RBC)** surface.
- There is **no blending**; both antigens appear **side by side**.

A person with blood group **AB** produces **both A and B antigens** on RBCs due to **co-dominant expression** of IA and IB alleles.

Human Blood Group Systems

Overview:

- Human blood group systems are **genetically determined** and based on **molecules (usually glycoproteins)** present on the **surface of red blood cells (RBCs)**.
- These systems are important for **safe blood transfusions** and **organ transplants**.

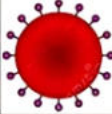
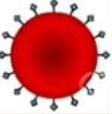
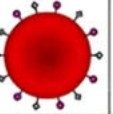
Key Systems:

Blood Group System	Key Molecules	Importance
ABO System	A and B antigens (glycoproteins)	Most widely used; determines A, B, AB, O
Rh System	Rh factor (D antigen)	Important in pregnancy and transfusions
MN System	M and N antigens	Rare; used in paternity testing

- ABO and Rh systems are the most significant due to their role in blood compatibility.
- The International Society of Blood Transfusion (ISBT) has recognized over 30 blood group systems.

Importance in Transfusions:

- Compatibility between **donor and recipient** is crucial.
- Incompatibility can trigger **immune reactions**, which may lead to **serious complications** or **death**.

Group	A	B	AB	O
Red Blood Cell Type				
Antigens Present	 Antigen A	 Antigen B	 Antigen A & B	None
Antibodies Present	 Anti-B	 Anti-A	None	 Anti-A & Anti-B





Summary:

- **Co-dominance in blood group AB** shows how both alleles can express **without masking** each other.
- **ABO and Rh systems** are critical in medical practices due to their impact on **transfusion safety**.

Multiple Alleles of ABO Blood Group System

Overview:

- The **ABO blood group system** is an example of a trait controlled by **multiple alleles**.
- There are **three alleles** involved: **IA**, **IB**, and **i**.
- The term "**I**" stands for **isoagglutinogen**, which is another word for **antigen**.

Function of Each Allele:

Allele	Antigen Produced	Dominance
IA	Antigen A	Dominant over I
IB	Antigen B	Dominant over I
i	No antigen (null expression)	Recessive to both IA and IB

- **IA and IB are co-dominant:** when both are present (IAIB), **both antigens** are produced → **AB blood group**.
- **i is recessive:** it only shows its effect when inherited in **homozygous** form (ii) → **O blood group**.

Genotypes and Phenotypes:

Blood Group	Possible Genotypes
A	IAIA or IAi
B	IBIB or IBi
AB	IAIB
O	ii

Expression Pattern:

- These alleles begin their expression **early in embryonic development**.
- The **blood group remains unchanged throughout life**, as the genes continue expressing **permanently**.

Key Points:

- The ABO system is a **classical example** of both **multiple alleles** and **co-dominance**.
- Understanding this system is crucial for **safe blood transfusions, organ transplants, and genetic studies**.





Blood Group Systems Based on RBC Antigens

Blood group system		Antigen on surface protein
1.	ABO	A ¹ A ² A ³ A ⁴ B and others
2.	Rhesus (R ^h)	D, C, c C ^w , Cx, E, D4 and other
3.	MNS	M, N, S, s
4.	P	P, P ⁱ and P ^k
5.	Lutheran	Lu ^a , Lu ^b
6.	Kell	K, k, K ^a , K ^h , Js ^a , Js ^b
7.	Lewis	Le ^a , Le ^b
8.	Duffy	Fy ^a , Fy ^b
9.	Diego	Di ^a , Di ^b
10.	Yt	Yt ^a , Yt ^b
11.	I	I, I
12.	Xg	Xg ^a
13.	Kidd	KJ

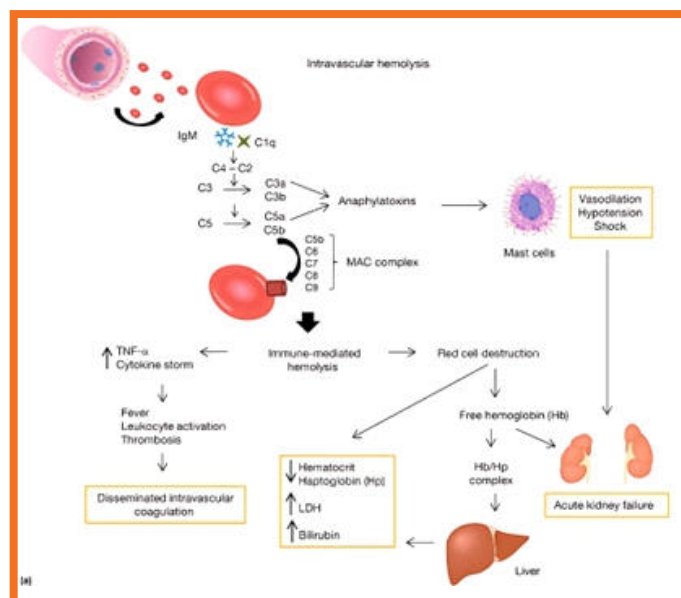
Blood Transfusion and Agglutination

- **Agglutination:** When blood of incompatible types is transfused, RBCs clump together, a process called agglutination, which is dangerous as it can block capillaries and impair circulation.
- Cross-matching: Before a transfusion, blood samples from both donor and recipient are cross-matched to check for compatibility.

Blood Group Antibodies and Transfusion Reactions

- **A group:** Contains Anti-B antibodies. If transfused with B blood, agglutination occurs.
- **B group:** Contains Anti-A antibodies. If transfused with A blood, agglutination occurs.
- **AB group:** Contains no Anti-A or Anti-B antibodies, can receive any blood type but can only donate to AB.
- **O group:** Contains both Anti-A and Anti-B antibodies, can receive only O blood but can donate to all blood types.





Hemolytic Reaction

- **Hemolytic reaction:** Occurs if incompatible blood is transfused. The recipient's antibodies destroy the donor's RBCs or vice versa, leading to potentially fatal consequences.

Key Takeaways:

- Compatibility is crucial in blood transfusions to prevent agglutination and hemolytic reactions.
- O blood group is the universal donor (for RBC transfusion), while AB is the universal recipient.

Negative (O-ve) as the Universal Donor

- **O Negative Blood Group Characteristics:**
- **Antigens:** O-ve individuals have neither A nor B antigens on their RBCs. Additionally, the Rh antigen is also absent.
- **Antibodies:** People with O-ve blood produce **anti-A** and **anti-B** antibodies in their serum.
- **Secretors:** Some individuals have a **secretor gene (Se)** on chromosome 19, meaning they secrete A and B antigens into their saliva and other body fluids. These individuals are known as secretors, but the absence of these antigens on their RBCs means their blood does not trigger an immune response in the recipient.

Why O Negative is the Universal Donor?

- Since O-ve blood has neither A, B, nor Rh antigens, it is not recognized by the recipient's immune system, thus reducing the chance of agglutination or hemolytic reactions when transfused to any recipient.
- The **absence of A, B, and Rh antigens** makes it compatible with all blood types, hence it is considered the universal donor for RBC transfusions.

AB Positive (AB+) as the Universal Acceptor

- **AB Positive Blood Group Characteristics:**
- **Antigens:** AB+ individuals have A, B, and Rh antigens on their RBCs.
- **Antibodies:** They do not produce anti-A or anti-B antibodies, meaning they do not reject transfusions from other blood groups.



Why AB Positive is the Universal Acceptor?"

- Since AB+ individuals have both A and B antigens, and the Rh antigen, their immune system will not react against any of the possible antigens found in other blood groups (A, B, AB, or O, with both positive and negative Rh factors).
 - As a result, AB+ individuals can safely receive blood from anyone—whether A, B, AB, or O, and regardless of Rh factor.

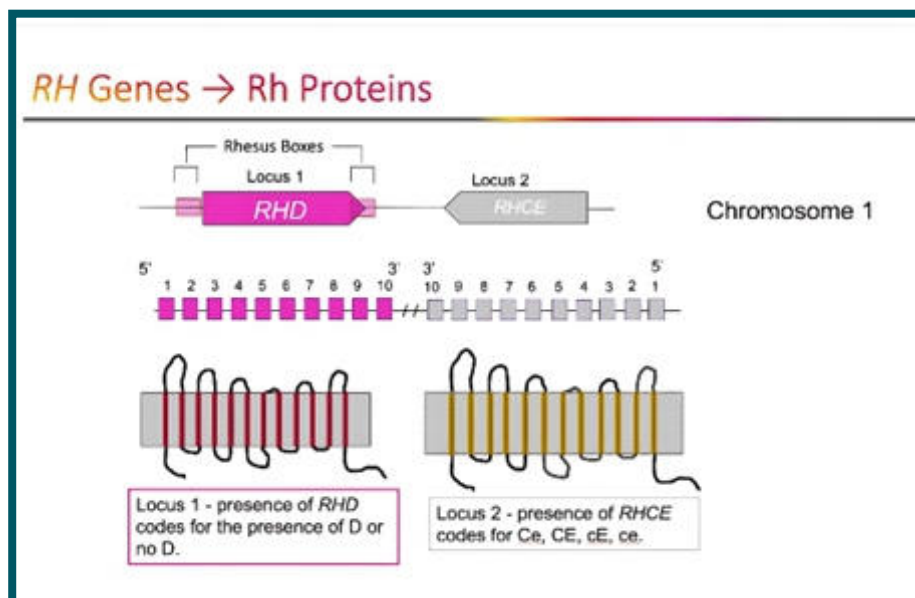
Rh Blood Group System

Overview:

- The **Rh blood system** is separate from the ABO blood group system but is equally important in blood transfusions and compatibility.
 - The Rh factor is represented by either **positive (+ve)** or **negative (-ve)**, which refers to the presence or absence of the **Rh antigen** (also called Rh factor) on the surface of RBCs.
 - This system is based on the **presence of the Rh antigen** on the RBCs, which was first observed in **Rhesus monkeys** by Landsteiner in the 1930s.

Rh Antigen and Gene Locus:

- The **Rh antigen** is primarily controlled by the **D gene locus**.
- There are two alleles at the Rh gene locus:
 - **D allele** (dominant)
 - **d allele** (recessive)
- **DD** or **Dd** genotypes will express the **Rh antigen** and be classified as **Rh positive (Rh+)**.
- **dd** genotype will not express the Rh antigen and will be classified as **Rh negative (Rh-)**.



Rh Positive (Rh+) and Rh Negative (Rh-) Individuals:

- **Rh+** individuals have the Rh antigen on their RBCs, meaning they produce Rh antigens and can be transfused with Rh+ or Rh- blood.
- **Rh-** individuals do not have the Rh antigen on their RBCs and cannot produce Rh antigens on their own.





Rh FACTOR				
Rh POSITIVE				
Rh NEGATIVE				

Antibody Production in Rh System:

- Unlike the ABO system, where **anti-A** and **anti-B** antibodies are naturally present, **anti-Rh antibodies** are not naturally occurring.
- **Anti-Rh antibodies** are produced only when an Rh- person is exposed to Rh+ blood (e.g., through transfusion or pregnancy), which stimulates the immune system to produce antibodies against the Rh antigen.

Transfusion Reactions:

- If an **Rh-** individual receives **Rh+** blood, their immune system will recognize the Rh antigen as foreign and produce **anti-Rh antibodies**.
- This leads to an **immune response**, where the recipient's body attacks and destroys the Rh+ RBCs from the donor, causing **agglutination** and **hemolysis** (destruction of RBCs).
- On the other hand, **Rh-** blood can be safely transfused to **Rh+** individuals since Rh- blood does not contain Rh antigens, and thus no immune response is triggered.

Rh Factor in Pregnancy:

- The Rh factor is especially significant in pregnancy. If an Rh- woman is carrying an Rh+ fetus (from an Rh+ father), she may develop **anti-Rh antibodies** during childbirth if Rh+ blood from the baby mixes with her blood.
- This could lead to **hemolytic disease of the newborn (HDN)** in subsequent pregnancies if the fetus is Rh+. To prevent this, **Rh- mothers** may receive an injection of **Rh immunoglobulin (RhIg)** to prevent the development of anti-Rh antibodies.

Key Points:

- **Rh+ (positive)**: Presence of Rh antigen (D allele dominant, DD or Dd genotype).
- **Rh- (negative)**: Absence of Rh antigen (dd genotype).
- **Rh+ blood** can be transfused to **Rh+** recipients.
- **Rh- blood** can be safely transfused to both **Rh+** and **Rh-** recipients.
- **Anti-Rh antibodies** are produced only after exposure to Rh+ blood and can cause serious reactions in future transfusions or pregnancies.

Erythroblastosis Fetalis

Overview:

- **Erythroblastosis fetalis** is a condition that arises due to **Rh incompatibility** between a mother and her fetus. It occurs when an **Rh-negative** mother carries an **Rh-positive** fetus, leading to serious complications for the newborn.





Cause:

- The condition occurs when an **Rh-negative mother** conceives a child with **Rh-positive blood** (inherited from the father).
- If the father has **DD genotype**, all children will be **Rh-positive (Dd)**.
- If the father has **Dd genotype** (heterozygous), there is a 50% chance of having **Rh+** or **Rh-** children.

Pathogenesis (Mechanism):

1. First Pregnancy:

- When an **Rh-negative mother** carries an **Rh-positive fetus**, some of the **Rh-positive fetal blood cells** may leak into the mother's bloodstream during childbirth.
- This triggers the **mother's immune system** to produce **anti-Rh antibodies** against the Rh antigen.

2. Subsequent Pregnancies:

- If the mother becomes pregnant again with another **Rh-positive fetus**, her **anti-Rh antibodies** can cross the placenta and attack the fetal **Rh-positive RBCs**.
- This causes the fetal RBCs to break down (hemolysis), leading to **anemia** in fetus.

3. Consequences of Hemolysis:

- The fetus becomes **anemic**, and the body responds by releasing **immature RBCs** (erythroblasts) into the bloodstream.
- This results in **erythroblastosis fetalis**, which is characterized by severe **hemolytic anemia** and the presence of **immature RBCs**.
- The liver and spleen of the fetus become enlarged due to the increased destruction of RBCs.

4. Bilirubin Accumulation:

- The breakdown of RBCs releases large amounts of **bilirubin**, a pigment that can accumulate in the fetus and cause **neonatal jaundice**.
- The bilirubin can damage the fetus's **nervous system**, leading to neurological impairment.
- The yellowing of the skin and eyes (jaundice) is a visible symptom of **bilirubin buildup**.

Potential Outcomes:

- **Abortion or stillbirth:** If severe hemolysis and anemia occur, it may lead to pregnancy loss.
- **Severe Jaundice:** The surviving baby may experience severe jaundice, and without intervention, it could cause long-term neurological damage.
- **Enlarged liver and spleen:** Due to the excessive breakdown of RBCs.

Preventive Measures:

- **Rh Antiserum (Rhlg):**
- To prevent the development of maternal anti-Rh antibodies in Rh-negative mothers, they are given **Rh antiserum (Rh immunoglobulin, Rhlg)** during early pregnancy and immediately after delivery.
- The Rh antibodies in the Rhlg injection bind to the fetal Rh-positive RBCs in the mother's bloodstream, preventing her immune system from recognizing them and producing antibodies.
- Rhlg helps prevent the development of **maternal antibodies** could affect future pregnancies.





- **Timing of Rhlg Injection:**

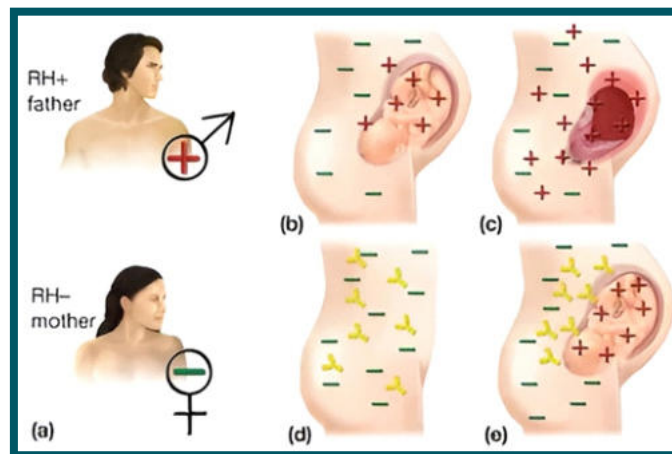
- The Rhlg injection is given during **early pregnancy** and again **immediately after delivery**.

- If the mother becomes pregnant again, the process should be repeated, as the antiserum only lasts for the duration of the pregnancy.

Treatment of Affected Newborns:

- If a baby is born with erythroblastosis fetalis and has high levels of bilirubin, the baby's blood may need to be replaced with Rh-negative blood (which does not contain anti-Rh antibodies) to prevent further hemolysis.

- This exchange transfusion helps remove the excess bilirubin and stabilize the baby.



Summary:

- **Erythroblastosis fetalis** is caused by **Rh incompatibility** between an Rh-negative mother and an Rh-positive fetus.

- **Maternal anti-Rh antibodies** attack the fetal RBCs, leading to **hemolytic anemia, jaundice**, and potentially neurological damage.

- **Rhlg (Rh immunoglobulin)** is given to Rh-negative mothers to prevent the formation of anti-Rh antibodies in future pregnancies.

- **Treatment** for affected babies may include an **exchange transfusion** to remove excess bilirubin.

Polygenic Inheritance and Epistasis

1. Polygenic Inheritance:

- **Definition:** Polygenic inheritance refers to the phenomenon where a **single trait** is controlled by **two or more genes** located at different loci. The combined effects of these multiple genes result in a **continuous range of phenotypic variation**.

- **Characteristics:**

- Traits controlled by multiple genes exhibit **quantitative inheritance**, meaning the phenotype shows **gradual variation** rather than discrete categories.

- The expression of the trait depends on the **additive effect** of each gene involved.

- **Examples:** Traits such as **height, skin color, weight**, and **intelligence** are influenced by many genes, each contributing to the final phenotype.

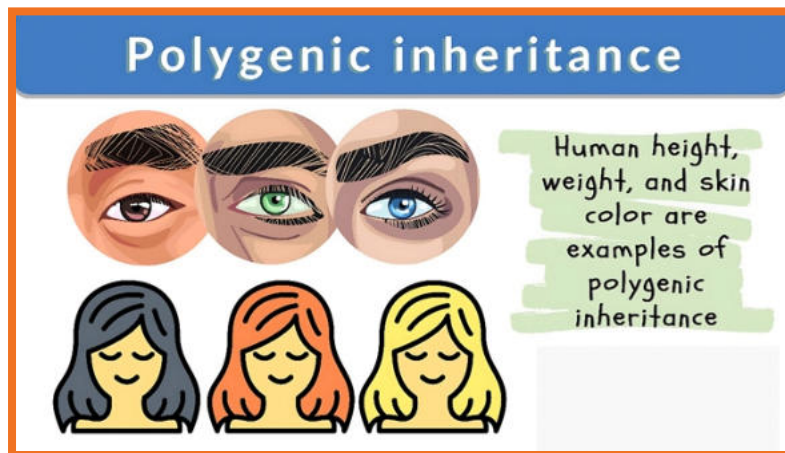
- **Continuously Varying Traits:** In polygenic inheritance, the phenotype can range smoothly between extremes, and individuals can exhibit a wide range of intermediate characteristics.





- **Key Points:**

- Polygenic traits are **quantitative**, meaning they can be measured on a spectrum (e.g., light to dark skin color).
- The effects of each gene are **additive** (they combine together to influence the phenotype).



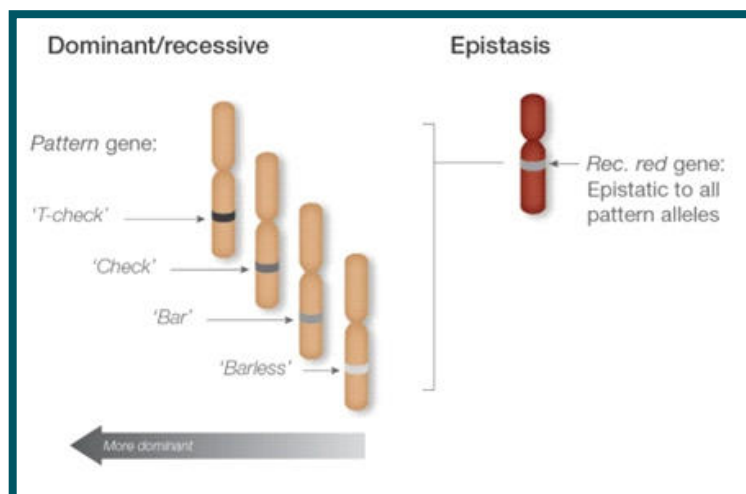
2. Epistasis:

- **Definition:** Epistasis is the phenomenon where the expression of one gene is **influenced or masked** by the presence of one or more other genes located at **different loci**. This gene interaction can modify or suppress the expression of the original gene.
- **Origin of the Term:** The word **epistasis** comes from the Greek words “epi” (above) and “stasis” (standing or stop), meaning one gene stands above and influences the expression of another gene.
- **Characteristics:**
 - Epistasis involves interactions between **different genes** that are located at **different loci**, whereas **dominance** refers to the relationship between different alleles of the same gene located at the same locus.
 - **Masking Effect:** One gene can mask or override the effect of another gene. The gene whose effect is masked is said to be **epistatic**, while the gene whose effect is masked is called the **hypostatic** gene.
- **Example:**
 - **Coat color in mice:** In mice, the gene for coat color (B) is epistatic to the gene for coat color expression (A). If a mouse has two recessive alleles for the epistatic gene (ee), the coat color gene will not be expressed, regardless of whether the mouse carries a dominant or recessive allele for coat color.
 - **Human Example:** In humans, the expression of the **albinism gene** (which causes a lack of pigment) can **mask** the expression of other genes controlling skin or eye color.

- **Key Points:**

- Epistasis involves **interactions between different genes** located at different loci.
- Dominance refers to the relationship between alleles of the same gene, while **epistasis** refers to the influence of one gene over the expression of another gene.
- Epistasis can result in **modified or suppressed phenotypes**, making it different from simple Mendelian inheritance.





Summary:

- **Polygenic Inheritance** involves the action of multiple genes contributing additively to a trait, resulting in **quantitative, continuously varying traits** (e.g., height, skin color).
- **Epistasis** is the interaction between genes at different loci, where one gene can **mask or modify** the expression of another gene. This is different from **dominance**, which involves interactions between alleles of the same gene.

Inheritance of Human Skin Color

1. Overview:

- Human skin color is a **polygenic trait** that shows continuous variation, meaning it can be found in a range of shades from very light to very dark.
- The variation in skin color is mainly due to the **amount and type of melanin** produced in the skin. Melanin is a pigment responsible for skin, eye, and hair color.
- The **amount of melanin** produced is influenced by the presence and interaction of multiple genes. The more melanin produced, the **darker the skin**.

2. Genes Involved in Skin Color:

- Skin color is influenced by **three to six gene pairs**. The more melanin-producing genes a person has, the **darker the skin color**.
- The primary genes involved are:
- **Gene A:** Responsible for the **survival, proliferation, and migration** of melanocytes (cells that produce melanin).
- **Gene B:** Controls the synthesis of **tyrosinase**, an enzyme that converts tyrosine into melanin.
- **Gene C:** Determines the type of melanin produced—**eumelanin** (dark brown/black) or **pheomelanin** (red/brown).

3. Alleles and Their Expression:

- The alleles for each gene come in **two forms**:
- **Dark skin alleles** (A, B, C) are represented by **capital letters**.
- **Light skin alleles** (a, b, c) are represented by **lowercase letters**.
- These alleles exhibit **incomplete dominance**:
- In **heterozygous condition**, the alleles result in an **intermediate phenotype** (i.e., neither fully dark nor fully light skin).
- **Homozygous dominant** (e.g., AABBCC) leads to darker skin.
- **Homozygous recessive** (e.g., aabbcc) leads to lighter skin.





4. Skin Color Spectrum:

- There are **seven different shades** of skin color ranging from **very light** (aabbcc) to **very dark** (AABCC).
- Most individuals have an **intermediate skin color** (e.g., AaBbCc), representing a mixture of alleles from both parents.
- An offspring of two parents with intermediate skin color (i.e., **mulatto** offspring) will also exhibit intermediate skin color, often the result of a cross between **black** and **white** parents.

5. Genetics of Skin Color:

- When two individuals with intermediate skin color (AaBbCc) cross, each parent can produce **8 different types of gametes**. These gametes will combine in various ways to produce **64 possible zygotes** (offspring), each with potentially **7 different skin colors**. The genetic combinations follow a **Mendelian inheritance pattern** with multiple gene loci influencing the final skin color.

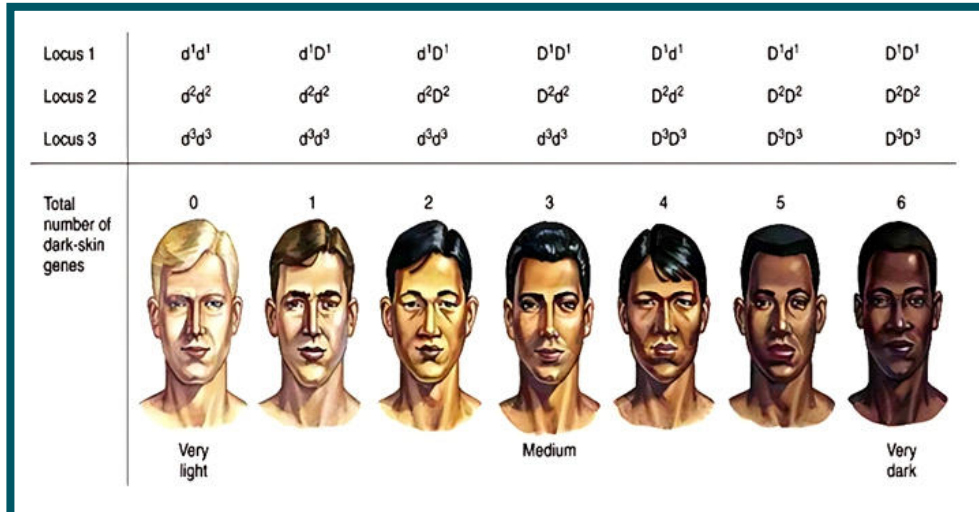
6. Summary:

Skin color in humans is a **polygenic trait** controlled by multiple genes that affect melanin production and type.

Three main genes (A, B, C) are involved, with their respective alleles controlling various aspects of melanin production.

The phenotype is influenced by the **combination of these alleles**, with the **dark skin alleles** contributing to more melanin production.

Skin color exhibits **continuous variation**, with **intermediate phenotypes** commonly seen in offspring of mixed ancestry.



Epistasis

1. Definition:

- Epistasis refers to the interaction between genes at different loci where one gene (the epistatic gene) interferes with or masks the expression of another gene (the hypostatic gene).
- Epistatic gene:** The gene that masks or interferes with the phenotype of another gene.
- Hypostatic gene:** The gene whose expression is masked by the epistatic gene.





2. Examples of Epistasis:

Foxgloves Petal Color:

- The color of the **foxglove plant** petals is controlled by **three genes**:

1. Gene M (for pigment synthesis):

- M (MM, Mm): Produces anthocyanin, resulting in **purple petals**.
- mm: Does not produce pigment, resulting in **white petals with yellowish spots**.

2. Gene D (for enhancer of anthocyanin):

- D (DD, Dd): Enhances the production of anthocyanin, resulting in **darker petals**.
- dd: Does not enhance pigment production, resulting in **lighter petals**.

3. Gene W (for color deposition):

- W (WW, Ww): Prevents pigment deposition except in spots, resulting in **white-spotted petals**.
 - ww: Allows pigment deposition in all cells, resulting in **uniformly colored petals**.
 - The combination of these genes produces different phenotypes:
- M (purple pigment) + D (enhancer) → Darker purple petals.**
 - m (no pigment) + W (white spots) → White-spotted petals.**
 - w (uniform color) → Uniform color petals.**

3. Phenotype Ratios in Di-hybrid Cross:

- In a **di-hybrid** cross where both genes are heterozygous, the offspring display a modified phenotype ratio due to **epistasis**:
- Without Epistasis:** If there were no epistasis, the expected ratio would be 9:3:3:1.
- With Epistasis:** Due to the interaction between genes W and D, the phenotype ratio changes to **12:3:1**.
- Phenotypes:
- White-spotted petals:** Produced by the combination of **ww** (epistasis gene) and other allelic combinations.
- Dark red petals:** Result from the presence of **D** and **M** alleles with the presence of **W**.
- Light red petals:** Result from the presence of **D** and **m** alleles, with **W**.

Punnett Square of Epistasis in Foxgloves:

Gametes	DW	Dw	dW	dw
DW	DDWW	DDWw	DdWW	DdWw
Dw	DDWW	DDWw	DdWW	DdWw
dW	DdWW	DdWw	ddWW	ddWw
dw	DdWW	DdWw	ddWW	ddWw





Results:

- The **epistasis gene (W)** prevents pigment deposition in the petal, except in the spots, leading to the **white-spotted** phenotype.
- D allele enhances the synthesis of **anthocyanin** (red pigment), and the **d allele** produces less pigment.

4. Dominant Epistasis in Foxgloves:

- Dominant Epistasis** refers to the **W allele** that prevents pigment deposition, overriding the effect of the other genes.
- The resulting **phenotype ratio** in the offspring of the di-hybrid cross (both parents heterozygous) is **12:3:1**:
- 12:** White-spotted petals (due to **ww** interaction with other genes).
- 3:** Dark red petals (due to **DD** or **Dd** and **M** with **W**).
- 1:** Light red petals (due to **dd** with **M** and **W**).

Gametes		DdWw				x	DdWw			
		DW	Dw	dW	dw	x	DW	Dw	dW	dw
		DW					Dw			
		dW					dw			
DW		DDWW	DDWw	DdWW	DdWw		DDWW	DDWw	DdWW	DdWw
		White spotted	White spotted	White spotted	White spotted		White spotted	White spotted	White spotted	White spotted
Dw		DDWw	DDww	DdWw	Ddww		DDWw	DDww	DdWw	Ddww
		White spotted	Dark red	White spotted	Dark red		White spotted	Dark red	White spotted	Dark red
dW		DdWW	DdWw	ddWW	ddWw		DdWW	DdWw	ddWW	ddWw
		White spotted	White spotted	White spotted	White spotted		White spotted	White spotted	White spotted	White spotted
dw		DdWw	Ddww	ddWw	ddww		DdWw	Ddww	ddWw	ddww
		White spotted	Dark red	White spotted	light red		White spotted	Dark red	White spotted	light red
Dominant Epistasis: white spotted : Dark red : light red										
12 : 3 : 1										

5. Summary:

- Epistasis** is the interaction between genes at different loci, where one gene masks or modifies the expression of another gene.
- In **foxgloves**, **epistasis** alters the typical **9:3:3:1** ratio to a **12:3:1** ratio in a di-hybrid cross.
- W** gene (epistatic) prevents pigment deposition in petals, resulting in **white-spotted** flowers.
- The **D gene** enhances pigment production, leading to darker red flowers.
- Epistasis** is important in determining complex traits like petal color in plants and plays a significant role in inheritance patterns.

1. Crossing Over:

- Linkage** refers to the tendency of genes located on the same chromosome to be inherited together. However, this is not always absolute.
- If **linkage** remains complete and no exchange of genetic material occurs, offspring will inherit traits exactly like one of the parents.
- During **meiosis**, homologous chromosomes pair up, and these pairs can undergo a process called **crossing over**, which results in the exchange of genetic material between chromatids.





2. Synapsis and Crossing Over:

- During **meiosis**, homologous chromosomes align and pair up in a process called **synapsis**.
- As the chromosomes align, they form **chiasmata**, which are sites where non-sister chromatids of homologous chromosomes exchange segments of their DNA. This process is known as **crossing over**.
- **Crossing over** leads to the recombination of alleles between chromosomes, resulting in new combinations of genetic material.

3. Outcome of Crossing Over:

- After crossing over, chromosomes carry new combinations of alleles, which increases genetic diversity. This results in **recombinant chromosomes**, which carry a mix of parental alleles that were not present in either parent.
- The genetic diversity created by crossing over leads to **genetic variability** during sexual reproduction, as it creates new combinations of genes in the offspring.

4. Example of Crossing Over:

- Consider two homologous chromosomes where:
- One chromosome carries the alleles **T** and **B**.
- The other chromosome carries the alleles **t** and **b**.
- **Chiasmata** form between the non-sister chromatids, and **crossing over** occurs, where:
- **Allele b** crosses over to the chromosome containing allele **T**.
- **Allele B** crosses over to the chromosome containing allele **t**.
- After crossing over, the chromatids separate into four possible combinations:
- **TB** (parental combination).
- **tb** (parental combination).
- **Tb** (recombinant combination).
- **tB** (recombinant combination).
- If crossing over did not occur, only the **parental combinations** (TB and tb) would be formed, resulting in no recombination of genetic material.

5. Significance of Crossing Over:

- Crossing over introduces **genetic variation** by producing **recombinant** chromosomes.
- It allows for the **shuffling** of genetic material, ensuring that offspring inherit a unique combination of genes.
- This process is a major contributor to **genetic diversity** in sexually reproducing organisms.

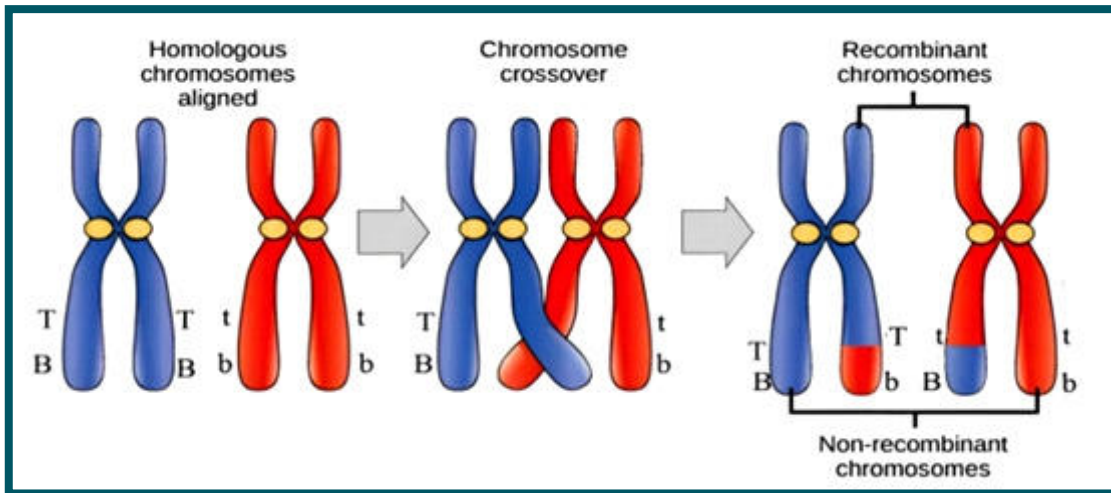
6. Non-recombinant and Recombinant Chromosomes:

- **Non-recombinant chromosomes:** Chromosomes that do not undergo crossing over and thus carry the same genetic combinations as the parent chromosomes.
- **Recombinant chromosomes:** Chromosomes that have undergone crossing over and contain a new combination of alleles, differing from the parental chromosomes.

7. Conclusion:

- Crossing over is a crucial mechanism that promotes **genetic variability** and increases diversity in offspring.
- By exchanging genetic material between chromatids, crossing over ensures that each gamete carries a unique combination of alleles, which is important for evolution and adaptation.





Sex Determination

Sex determination refers to the process by which the biological sex of an individual is established. There are various systems of sex determination in different organisms, with some systems using chromosomes to determine the sex of the individual.

1. Chromosomal Basis of Sex Determination:

- **Autosomes:** Most chromosomes in animals and plants come in homologous pairs, meaning both members of a pair carry the same type of genes.
- **Sex Chromosomes:** A special pair of chromosomes determines the sex of the individual. These chromosomes are not homologous, meaning they are different from each other. The two main types of sex chromosomes are **X** and **Y**.

2. Sex Determination in Grasshoppers (W. Sutton's Discovery):

- **W. Sutton** observed that in grasshoppers, the number of sex chromosomes is different in males and females.
- **Females** have **24 chromosomes** (12 pairs of chromosomes), with both autosomes and sex chromosomes being homologous. Each egg carries **12 chromosomes**.

- **Males** have **23 chromosomes** (11 pairs of autosomes and one sex chromosome).

Their sperm can carry two types of sex chromosomes:

- One type of sperm carries **11 chromosomes**.
- The other type of sperm carries **12 chromosomes**.
- **Fertilization Outcome:**
 - If a sperm with **11 chromosomes** fertilizes the egg, a male offspring is produced.
 - If a sperm with **12 chromosomes** fertilizes the egg, a female offspring is produced.
- **Sex Chromosome:** The odd chromosome (which can be the extra chromosome) determines the sex of the individual. These chromosomes are called **sex chromosomes**, while the other chromosomes are known as **autosomes**.

3. XX and XY Sex-Determination in Drosophila and Mammals:

- **Drosophila melanogaster** (Fruit Fly), studied by **T.H. Morgan**:
 - Morgan observed that the **male Drosophila** has one **X chromosome** and one **Y chromosome**, while the **female Drosophila** has **two X chromosomes**.
 - This XY system of sex determination is similar to that in humans and many other animals.





In *Drosophila*:

- **XX**: Female (two X chromosomes).
- **XY**: Male (one X and one Y chromosome).

4. Mechanism of Sex Determination in *Drosophila* and Mammals:

• **Males:**

• In **males**, the presence of a **Y chromosome** leads to male sex determination. The Y chromosome carries the **Sry gene** (Sex-determining Region Y), which is crucial for the development of male characteristics.

• **Females:**

• In **females**, the absence of the Y chromosome and the presence of two **X chromosomes** lead to the development of female characteristics.

5. Sex Chromosome Systems:

• The **XX-XY system** is the most common system of sex determination found in humans, many animals, and certain plants.

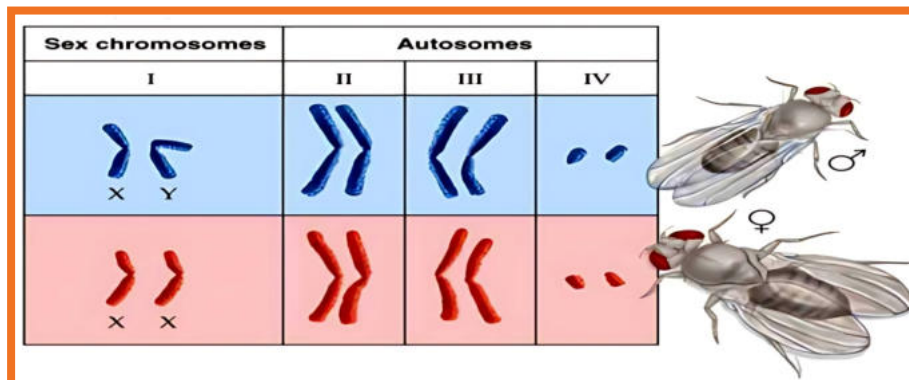
- **Males** have **XY** sex chromosomes (heterogametic).

- **Females** have **XX** sex chromosomes (homogametic).

6. Conclusion:

• The **chromosomal theory of inheritance** helps explain the inheritance of sex through sex chromosomes.

• Different organisms use different mechanisms of sex determination, but in many species, including humans and *Drosophila*, the presence or absence of sex chromosomes (X and Y) determines the biological sex of an individual.



Karyotype of *Drosophila* (Fruit Fly)

The karyotype of ***Drosophila melanogaster*** (the common fruit fly) provides important insights into sex determination and chromosome structure. The karyotype shows the number and structure of chromosomes in an organism's cells.

1. Chromosome Pairs:

• **Autosomes:** There are **three pairs of autosomes** (non-sex chromosomes) that are **identical** in both male and female flies. These autosomes carry the majority of the organism's genetic information.

• **Autosomes** are the chromosomes that are the same in males and females and do not determine sex.

2. Sex Chromosomes (4th Pair):

• **Sex Chromosomes:** The **4th pair of chromosomes** in *Drosophila* determines the sex of the individual.





- **Females** have two **identical rod-shaped** chromosomes in this pair, which are **homologous**. Both of these chromosomes are **X chromosomes**.
- **Males** have **two different chromosomes** in this pair. One is **rod-shaped (X)** and the other is **hook-shaped (Y)**. The presence of the Y chromosome determines the male sex.

3. Sex Determination System in Drosophila:

- **XX:** Female flies have **two X chromosomes (XX)**, which are homologous and identical in shape.
- **XY:** Male flies have one **X chromosome** and one **Y chromosome (XY)**. The Y chromosome is non-homologous to the X chromosome and has a distinct hook shape.

4. Summary of Chromosomal Structure in Drosophila:

- **Autosomes (3 pairs):** Identical in males and females.
- **Sex Chromosomes (4th pair):**
- **Female:** Two identical X chromosomes (XX).
- **Male:** One X chromosome and one Y chromosome (XY).

5. Sex Chromosomes and Their Role:

- **X Chromosome:** The X chromosome carries many essential genes. In females (XX), both X chromosomes carry genetic information that contributes to the organism's traits.
- **Y Chromosome:** The Y chromosome carries the **Sry gene**, which is responsible for the development of male characteristics in many organisms, including Drosophila.

Sex Determination in Humans

Human sex determination follows a system similar to that of **Drosophila**, but with differences in the structure of sex chromosomes. Here's an overview of how sex determination works in humans:

1. Chromosome Composition:

- **Humans** have **46 chromosomes** in total, which are arranged into **23 pairs**.
- **22 pairs** are **autosomes**, which are identical in both males and females. These autosomes carry most of the genetic information.
- The **23rd pair** are the **sex chromosomes**, which differ between males and females and determine the sex of the individual.

2. Sex Chromosomes:

- **Female (XX):** The **23rd pair** in females consists of two **identical X chromosomes (XX)**, which are homologous.
- **Male (XY):** The **23rd pair** in males consists of one **X chromosome** and one **smaller Y chromosome (XY)**. The Y chromosome is much smaller than the X chromosome and differs in shape.

3. X and Y Chromosomes:

- **X Chromosome:** The X chromosome carries many essential genes, and both males and females possess at least one X chromosome.
- **Females** inherit **two X chromosomes**.
- **Males** inherit one **X chromosome** from their mother.
- **Y Chromosome:** The Y chromosome contains the Sry gene, which triggers the development of male characteristics. It is present only in males, and its presence determines male sex.





4. Somatic and Germ Cells:

- **Somatic Cells:** These are the body cells in both males and females. They contain 46 **chromosomes** ($2n$), including **44 autosomes** and **2 sex chromosomes**.
- **Germ Cells:** These are the reproductive cells (sperm and egg). They contain **half the number of chromosomes**, i.e., **23 chromosomes** (n).
- **Females** produce eggs that contain only **X chromosomes**.
- **Males** produce sperm that can carry either an **X chromosome** or a **Y chromosome**.

5. Sex Determination Mechanism:

- **Female (XX):** A female produces only **X-carrying eggs**. When an **X-carrying sperm** fertilizes the egg, the resulting offspring will be **XX** (female).
- **Male (XY):** A male produces **X-carrying sperm** and **Y-carrying sperm**. If a **Y-carrying sperm** fertilizes the egg, the offspring will be **XY** (male). If an **X-carrying sperm** fertilizes the egg, the offspring will be **XX** (female).

6. Key Concepts:

- **Heterogametic Sex:** The **male** is considered **heterogametic** because he produces two types of sperm (X and Y).
- **Homogametic Sex:** The **female** is considered **homogametic** because she produces only one type of egg (X).

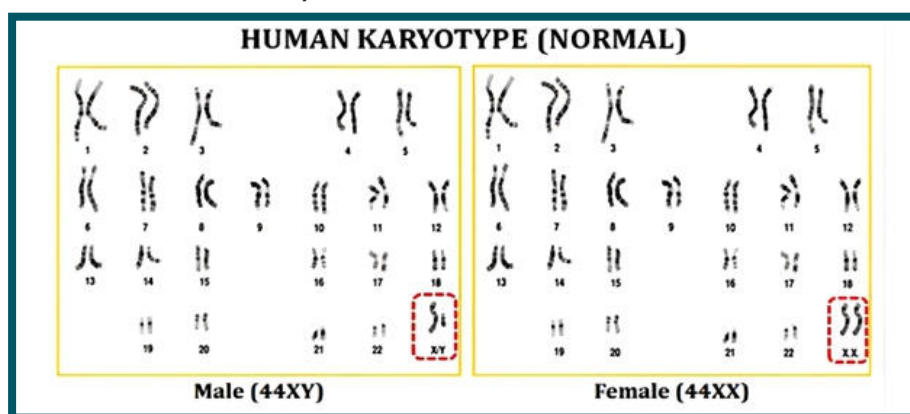
Karyotype in Humans:

- **Human Female Karyotype:** 44 autosomes + XX sex chromosomes.
- **Human Male Karyotype:** 44 autosomes + XY sex chromosomes.

7. Summary:

- **Human females** have two X chromosomes (XX), while **males** have one X and one Y chromosome (XY).
- **X chromosome** determines many traits and is present in both sexes, while the **Y chromosome** is responsible for male sex determination.

This system of sex determination ensures the diversity of sexes in humans and other organisms that follow the XX/XY system.



Sex Linkage Overview:

- According to the **chromosomal theory of inheritance**, genes are located on chromosomes.
- **Linked genes** are genes located on the same chromosome, and they tend to be inherited together unless crossing over occurs.
- If crossing over doesn't occur, the linked genes are inherited as a block.





Types of Sex Linkage:

1. Autosomal Linked Genes:

- These genes are located on **autosomes** (non-sex chromosomes).
- Their inheritance follows the **autosomal linked inheritance** pattern, where genes located on the same chromosome tend to be inherited together.

2. Sex-Linked Genes:

- These genes are located on sex chromosomes (X and Y).
- The two types of sex chromosomes are:
 - **X-chromosome:** Contains a large number of genes.
 - **Y-chromosome:** Contains fewer genes, primarily those that determine maleness.
- **X-Linked Inheritance:**
 - Traits controlled by genes on the **X-chromosome** are called **X-linked traits**.
 - The inheritance of these traits follows the **X-linked inheritance** pattern, with notable differences between male and female inheritance due to the presence of a single X in males (XY) and two Xs in females (XX).
- **Y-Linked Inheritance:**
 - Traits controlled by genes on the **Y-chromosome** are called **Y-linked traits**.
 - These traits are passed only from father to son because only males have a Y-chromosome.

3. X-Y Linked (Pseudo Autosomal Genes):

- Some traits are controlled by genes found on both **X** and **Y chromosomes**. These are called **X-Y linked genes** or **pseudo-autosomal genes**.
 - They have an **autosomal pattern of inheritance**, meaning they follow a pattern similar to autosomal inheritance, but the genes are located on sex chromosomes.

4. Genes on the Y Chromosome:

- While the Y chromosome is often considered inert in terms of genetic inheritance, it does carry some genes.
 - These Y-linked genes express traits related to maleness, such as the **bobbed gene** in **Drosophila**, which exhibits inheritance patterns similar to autosomal genes.

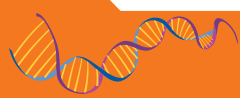
Key Concepts:

- **Sex-linked inheritance** refers to genes located on sex chromosomes (X or Y).
- **X-linked inheritance:** Traits controlled by genes on the X chromosome.
- **Y-linked inheritance:** Traits controlled by genes on the Y chromosome.
- Pseudo-autosomal genes are located on both X and Y chromosomes and have inheritance patterns similar to autosomal genes.
- **Y chromosome genes** primarily affect male traits and have no contrasting alleles on the X chromosome.

Sex-Linked Inheritance in Drosophila (Fruit Flies)

Background:

- **T.H. Morgan** (1910) conducted breeding experiments on **Drosophila melanogaster** (fruit fly) and observed a **white-eyed mutant** male in a population of wild-type red-eyed flies.
 - Morgan noted that the white-eyed trait seemed to follow a distinct pattern of inheritance that did not align with Mendelian ratios.





Initial Crosses:

- **Wild-type red-eyed female** (homozygous red-eyed) was crossed with a **white-eyed male** (mutant).
- **F1 generation:** All offspring had red eyes, indicating that the red-eye allele (R) was dominant over the white-eye allele (r).
- **F2 generation:** When the F1 individuals were crossed, the expected **Mendelian ratio** (3:1, red-eyed to white-eyed) was observed in the F2 generation.

Key Observations:

- When a **white-eyed female** was crossed with a **red-eyed male**, the results were not the same as the previous cross.
- This suggested that the **white-eye mutation** was linked to the **X chromosome** and was not autosomal.

Genetic Explanation:

- The **red-eye allele (R)** is **dominant** and located on the **X chromosome**.
- The **white-eye allele (r)** is **recessive** and also located on the **X chromosome**.

Results of Crosses:

- **Cross 1 (Red-eyed female × White-eyed male):**
 - Female: **XX** (homozygous red-eye).
 - Male: **XrY** (white-eyed mutant).
 - All F1 offspring (both male and female) had red eyes. Males inherited their **X chromosome** from their mother and their **Y chromosome** from their father, thus receiving the dominant red-eye allele from the mother.
- **Cross 2 (White-eyed female × Red-eyed male):**
 - Female: **XrXr** (white-eyed).
 - Male: **XY** (red-eyed).
 - In this cross, the results were different from the first cross:
 - **Females:** All F1 females inherited an Xr from their mother and an X from their father, so they all had red eyes (XRXr).
 - **Males:** All F1 males inherited an Xr from their mother and a Y from their father, resulting in white eyes (XrY).

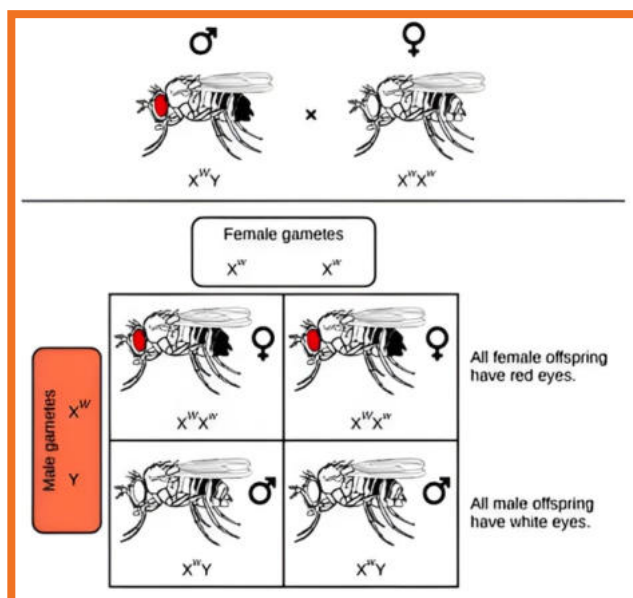
Conclusion:

- The **white-eye mutation** is **X-linked** and **recessive**.
- The **dominant red-eye allele** is located on the X chromosome, and males are more likely to express the recessive white-eye trait because they only have one X chromosome.
- **Females** can carry the white-eye allele but will only express it if they inherit the recessive allele from both parents (homozygous XrXr).

Sex-Linkage in Drosophila:

- In **Drosophila**, the inheritance of sex-linked traits follows a different pattern than autosomal traits because the genes for these traits are located on the **X chromosome**. The Y chromosome does not carry these alleles.
- **Male Drosophila** are **hemizygous** for the X chromosome (only one X), so they will express whatever allele they inherit on their single X chromosome, whether dominant or recessive.
- **Female Drosophila** have two X chromosomes, so they require two copies of the recessive allele for the trait to be expressed.





Cross-I			Cross-II		
Red eyed	X	white eyed	white eyed	X	Red eyed
Female		Male	Female		Male
$X^R X^R$		$X^r Y$	$X^r X^r$		$X^R Y$

Sex-Linkage Inheritance in Humans

Humans follow the same pattern of **autosomal and sex chromosome inheritance**:

- Females have **XX** sex chromosomes.
- Males have **XY** sex chromosomes.
- X-Linked Traits in Humans:**
- The **X chromosome** carries a variety of genes that control specific traits, including:
 - Color blindness** (recessive)
 - Muscular dystrophy** (recessive)
 - Hemophilia** (recessive)
 - Congenital deafness** (recessive)
 - Diabetes insipidus** (can be recessive or dominant)
- Inheritance Patterns:**
 - Recessive X-linked traits: These traits are typically more common in males, as they have only one X chromosome. A single recessive allele on the X chromosome will result in the trait being expressed in males.
 - Dominant X-linked traits: These traits can be expressed in both males and females, but the expression is more common in females since they have two X chromosomes.
- Genetic Studies in Humans:**
 - In human genetics, inheritance is often studied through pedigree analysis, as controlled breeding experiments are not feasible.
 - Pedigrees help trace the inheritance patterns of sex-linked and autosomal traits over generations.





X-linked Dominant Inheritance

- **Dominant X-linked Disorders:**

- In these disorders, the gene responsible for the disorder is located on the X chromosome and is dominant.

- **Female Inheritance:**

- A female can be affected if she carries one dominant allele (heterozygous) or two dominant alleles (homozygous).

- A heterozygous female has a 50% chance of passing the disorder to her offspring, regardless of the offspring's sex.

- **Male Inheritance:**

- If an affected father has the disorder, all his daughters will inherit the affected X chromosome and thus be affected by the disorder.

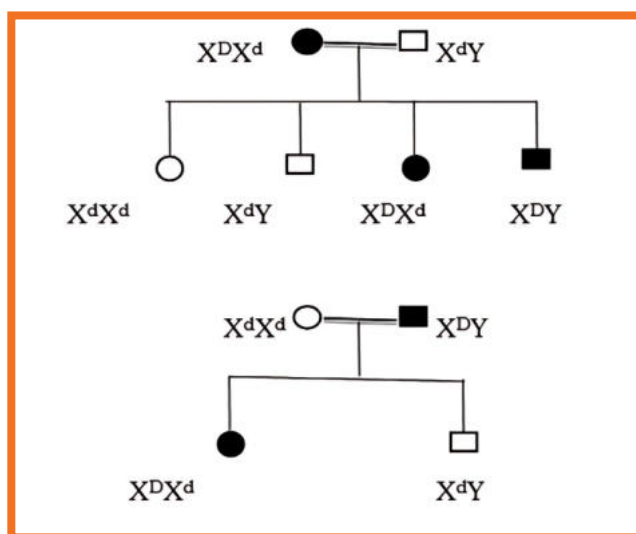
- None of the sons will inherit the disorder, as they inherit the Y chromosome from their father.

- **Examples of X-linked Dominant Disorders:**

- Vitamin D resistant rickets (hypophosphatemia)

- Rett syndrome (a neurological disorder affecting girls)

This pattern of inheritance shows how sex-linked traits are passed differently between males and females, with X-linked recessive traits being more common in males, and X-linked dominant traits potentially affecting both sexes but with different inheritance patterns.



Genetic Inheritance of Hemophilia

- **Definition:**

- **Hemophilia** is an **X-linked recessive** disorder where individuals have difficulty with blood clotting after injury. This is due to an inability to produce certain blood clotting factors.

- **Types of Hemophilia:**

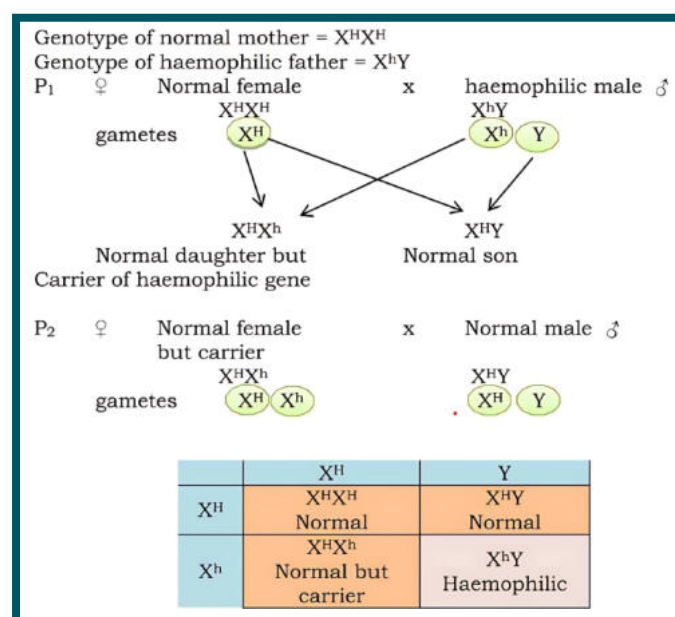
- **Hemophilia A and B:** Both are **X-linked recessive** traits.

- **Hemophilia C:** An **autosomal recessive** disorder, not linked to the X chromosome, and affects both males and females equally.





- **Why Hemophilia A and B are More Common in Males:**
- **Males** are **hemizygous** for the X chromosome (having only one X chromosome), so a single **recessive allele** on the X chromosome will cause hemophilia.
- **Females** have two X chromosomes and need to inherit two copies of the **recessive allele** (one from each parent) to develop hemophilia, making it less common in females.
- **Inheritance Pattern:**
- Hemophilia follows a **zig-zag inheritance pattern**:
- It is passed from a **maternal grandfather** (who carries the defective allele on his X chromosome) through a **carrier daughter** to a **grandson**.
- **Hemophilia does not pass directly from father to son** since males inherit the Y chromosome from their father, not an X chromosome.
- **Genotypic Example:**
- A **normal mother** has the genotype: $X^H X^H$ (homozygous dominant).
- A **hemophiliac father** has the genotype: $X^h Y$ (X-linked recessive allele for hemophilia).
- **Gametes and Offspring:**
- **Gametes of the normal mother ($X^H X^H$):** X^H or X^H .
- **Gametes of the hemophiliac father ($X^h Y$):** X^h or Y .
- **Possible offspring:**
- **Male offspring ($X^H Y$):** Normal, because they inherit the Y chromosome from their father and the X^H from their mother.
- **Female offspring ($X^H X^h$):** Carrier of the hemophilia gene, because they inherit one normal X chromosome (X^H) from their mother and one defective X chromosome (X^h) from their father.
- **Conclusion:**
- Hemophilia **cannot** pass directly from father to son.
- Sons inherit the **X chromosome** from their mother and **Y chromosome** from their father.
- **Carrier females** pass the hemophilia gene to their sons, but they will not show symptoms unless both X chromosomes carry the recessive allele.





Inheritance of Color Blindness

- **Color Vision Mechanism:**

- The human eye contains two types of neurons:
- **Rods:** Responsible for vision in low light.
- **Cones:** Responsible for color distinction.
- Humans typically have **trichromatic vision**, allowing the differentiation of three primary colors: **red, green, and blue**. These colors are detected by three types of cones in the retina, each sensitive to one primary color.

- **Opsins:**

- Each cone type contains **light-absorbing proteins** called **opsins**, which are produced by genes.
- The **genes for red and green opsins** are located on the **X-chromosome**, while the **gene for blue opsin** is located on **autosome chromosome 7**.

- **Mutations and Color Blindness:**

- Mutations in the opsin genes can lead to **recessive alleles** causing various forms of **color blindness**. These forms of color blindness are categorized as:

- **Dichromacy:** The person can distinguish two colors but cannot perceive one of the three primary colors due to a mutated opsin gene.

- **Monochromacy:** The person can recognize only one color, leading to true color blindness.

- **Types of Color Blindness:**

- **Dichromacy** (inability to perceive one primary color):
- **Protanopia: Red blindness** (cannot distinguish red).
- **Deuteranopia: Green blindness** (cannot distinguish green).
- **Tritanopia: Blue blindness** (cannot distinguish blue).
- **Monochromacy (total color blindness):**

- In **blue monochromacy**, the individual can only recognize blue but cannot perceive red or green.

- This type of color blindness occurs when both red and green cone opsins are absent.

- **Inheritance Pattern:**

- **X-linked recessive trait:** The gene mutations for color blindness are located on the X chromosome, making this condition more common in **males** than in **females**.

- Like hemophilia, the inheritance follows a **zig-zag (criss-cross)** pattern:

- Males inherit their X chromosome from their **mother**, and the defective gene on the X chromosome causes the condition.

- Females can only be affected if they inherit two copies of the mutated gene (one from each parent), making color blindness much rarer in females.

- **Symptoms:**

- Normal individuals will see all three primary colors clearly.
- Individuals with monochromacy will only see one color, usually blue, while they cannot distinguish red or green.

- **Example of Color Blindness Inheritance:**

- A **carrier mother ($X^H X^h$)** can pass the gene for color blindness to her sons, but the sons will have a 50% chance of being colorblind.





- A **father** with color blindness (X^hY) will pass the colorblind X chromosome to his daughters, making them carriers, but his sons will inherit his Y chromosome and will not be colorblind

Symbols of Pedigree Development

	Male
	Female
	Affected Male
	Affected Female
	Carrier Female for sex linked inheritance
	Heterozygous male for autosomal recessive trait.
	Heterozygous female for autosomal recessive trait
	Mating or cross
	Mating or cross
	Dizygotic twins
	Monozygotic twins

